

Points of intersection of lines and circles [Cambridge revision]

Substitute, solve the quadratic, and let the discriminant say how many points there are.

LESSON 57–58

2 × 40 MIN

REVISION OF GRADE 9

P1 3.5

LESSON CLOCK

15'

25'

25'

20'

15'

Two lines	15 min
A line and a circle	25 min
The discriminant decides	25 min
The tangent condition	20 min
Homework	15 min

LEARNING OBJECTIVES

- Find where two lines meet by solving their equations simultaneously.
- Find where a line meets a circle by substitution.
- Use the discriminant to count intersections without solving.
- Find the condition for a line to be a tangent to a circle.

TEXTBOOK

National	Geometry 9, pp. 133–146
Cambridge	Pure Mathematics 1, pp. 65–72
Workbook	Exercise 3E

01

Explanation

Two lines

Two lines meet where both equations hold. Solve simultaneously — by substitution when one equation is already $y = \dots$, by elimination otherwise.

PAIR	RELATION	MEETING POINTS
$y = 2x + 1, y = -x + 7$	different gradients	(2, 5)
$y = 2x + 1, y = 2x + 5$	parallel	none
$y = 2x + 1, 4x - 2y + 2 = 0$	the same line	infinitely many

THREE OUTCOMES, AND WHAT EACH LOOKS LIKE ALGEBRAICALLY

A unique solution, no solution (a contradiction such as $0 = 4$), or infinitely many (an identity such as $0 = 0$). The algebra always tells you which before the picture does.

A line and a circle

Make y the subject of the line, substitute into the circle, and a quadratic in x appears. Its roots are the x -coordinates of the meeting points.

Example. $y = x + 1$ and $x^2 + y^2 = 25$:

STEP	WORKING
substitute	$x^2 + (x + 1)^2 = 25$
expand	$2x^2 + 2x - 24 = 0$
simplify	$x^2 + x - 12 = 0$
solve	$x = 3, x = -4$
back-substitute	$(3, 4)$ and $(-4, -3)$

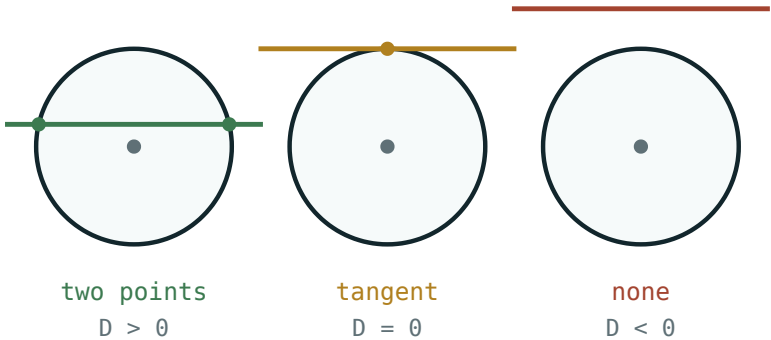
FINISH THE JOB – FIND Y TOO

A point has two coordinates. Stopping at $x = 3, x = -4$ answers half the question and earns half the marks. Substitute back into the **line**, which is easier than the circle.

The discriminant decides

The number of meeting points is the number of roots of that quadratic — so it is decided by $D = b^2 - 4ac$ alone.

$D > 0$ two points $D = 0$ tangent $D < 0$ no points



Three positions, three signs of the discriminant.

This lets you answer “how many times does the line meet the circle?” without solving anything — form the quadratic, compute D , stop.

LINE WITH $x^2 + y^2 = 25$	QUADRATIC	D	POINTS
$y = x + 1$	$x^2 + x - 12 = 0$	49	2
$y = x + 5\sqrt{2}$	$2x^2 + 10\sqrt{2}x + 25 = 0$	0	1
$y = x + 10$	$2x^2 + 20x + 75 = 0$	-200	0

The tangent condition

A line is a tangent exactly when $D = 0$. Setting $D = 0$ and solving for the unknown in the line gives every tangent of that family.

Example. For which c is $y = x + c$ a tangent to $x^2 + y^2 = 25$?

$$2x^2 + 2cx + c^2 - 25 = 0, D = 4c^2 - 8(c^2 - 25) = 200 - 4c^2$$

So $D = 0$ when $c^2 = 50$, that is $c = \pm 5\sqrt{2}$ — the two tangents parallel to $y = x$, one on each side.

THE GEOMETRIC ROUTE, OFTEN FASTER

A line is a tangent when the distance from the centre to the line equals r :

$$d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} = r$$

For $y = x + c$ and the centre $(0, 0)$: $\frac{|c|}{\sqrt{2}} = 5$, giving $c = \pm 5\sqrt{2}$ in one line.

Chord length. If $d < r$, the half-chord, d and r form a right triangle, so the chord is $2\sqrt{r^2 - d^2}$ — no quadratic needed.

02 Worked examples

EXAMPLE 1 Find where $y = x + 1$ meets $x^2 + y^2 = 25$.

① $x^2 + (x + 1)^2 = 25$

② $2x^2 + 2x - 24 = 0 \Rightarrow x^2 + x - 12 = 0$

③ $x = 3$ or $x = -4$

④ $(3, 4)$ and $(-4, -3)$

Back into the line.

ANSWER

$(3, 4)$ and $(-4, -3)$

EXAMPLE 2 How many times does $y = 2x + 8$ meet $x^2 + y^2 = 9$?

① $x^2 + (2x + 8)^2 = 9$

② $5x^2 + 32x + 55 = 0$

③ $D = 1024 - 1100 = -76$

④ Negative — the line misses the circle.

ANSWER

No points of intersection

EXAMPLE 3 For which c is $y = 2x + c$ a tangent to $x^2 + y^2 = 20$?

① Distance from $(0,0)$: $\frac{|c|}{\sqrt{5}}$

② $\frac{|c|}{\sqrt{5}} = \sqrt{20} = 2\sqrt{5}$

③ $|c| = 10$

④ $c = \pm 10$

ANSWER

$c = 10$ or $c = -10$

EXAMPLE 4 Find the length of the chord cut on $x^2 + y^2 = 25$ by $y = 3$.

① $d = 3$ from the centre.

② Half-chord $\sqrt{25 - 9} = 4$
Right triangle.

③ Chord = 8

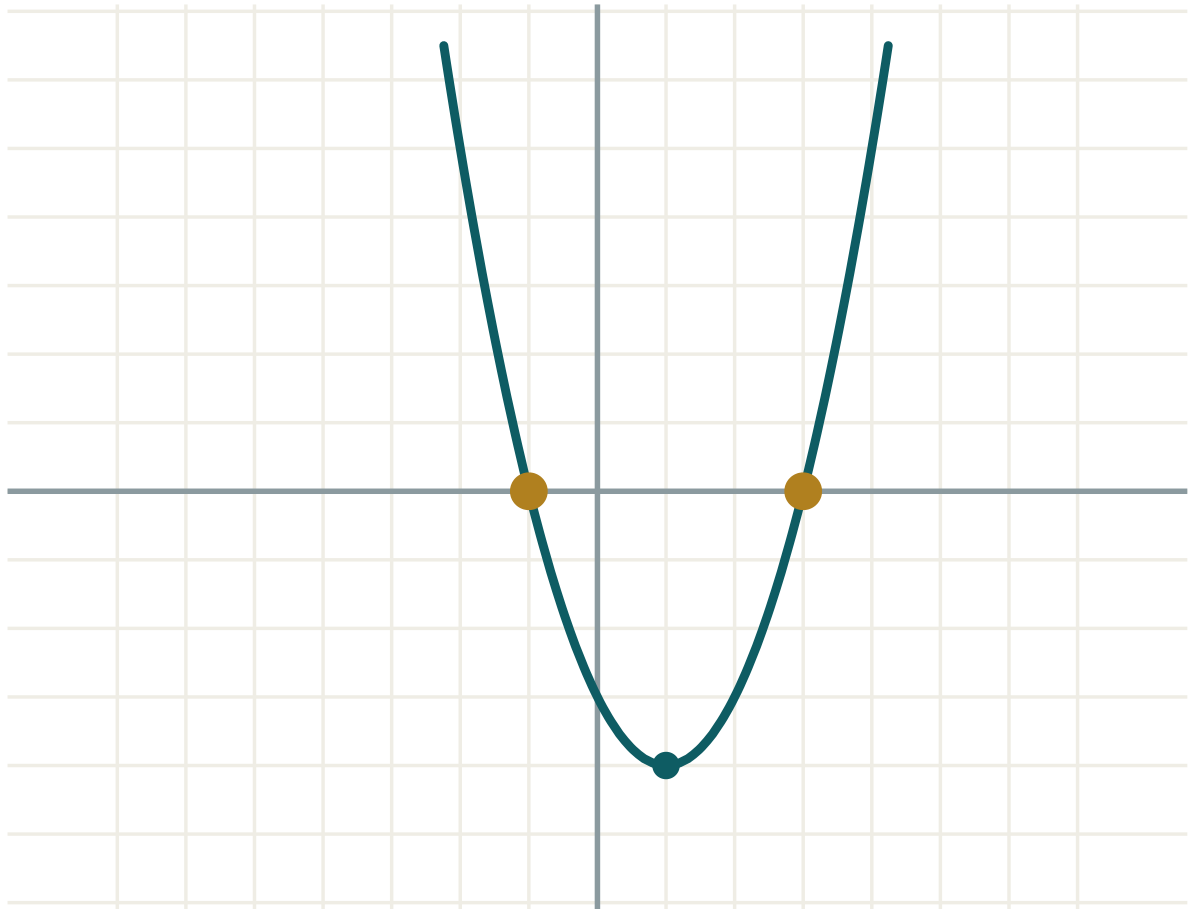
ANSWER

8

03 Interactive model

Draw one circle and slide a ruler across it, stopping at each of the three positions; name the sign of D each time.

The discriminant, and how many points

a **1**b **-2**c **-3** $D = b^2 - 4ac$ **16**roots **two** x_1 **-1** x_2 **3** $x_1 + x_2$ **2** $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Point of intersection	Kesishish nuqtasi	Точка пересечения
Simultaneous equations	Tenglamalar sistemasi	Система уравнений
Substitution	O'rniga qo'yish	Подстановка
Discriminant	Diskriminant	Дискриминант
Tangent condition	Urinma sharti	Условие касания
Secant	Kesuvchi	Секущая
Chord length	Vatar uzunligi	Длина хорды
Distance from a point to a line	Nuqtadan chiziqqa masofa	Расстояние от точки до прямой

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A line meets a circle in at most:

one point

two points

three

four

$D = 0$ means the line is:

a secant

a tangent

missing the circle

a diameter

After finding x you must:

stop

find y from the line

find y from the circle

square it

Two parallel lines give:

A line is a tangent when its distance from the centre is:

The chord at distance d from the centre has length:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Where do $y = 2x + 1$ and $y = -x + 7$ meet?

ANSWER $(2, 5)$

2. Where do $y = 3x$ and $y = 3x + 2$ meet?

ANSWER Nowhere — parallel

3. Does $(3, 4)$ lie on $x^2 + y^2 = 25$?

ANSWER Yes

4. D of $x^2 + x - 12 = 0$

ANSWER 49

5. Chord of $x^2 + y^2 = 25$ along $y = 3$

ANSWER 8

6. Chord of $x^2 + y^2 = 25$ along $y = 0$

ANSWER 10

7. Distance from $(0,0)$ to $y = x + 4$

ANSWER $\frac{4}{\sqrt{2}} = 2\sqrt{2}$

Medium · the standard question

7 PROBLEMS

1. Where does $y = x + 1$ meet $x^2 + y^2 = 25$?

ANSWER $(3, 4), (-4, -3)$

2. How many points has $y = 2x + 8$ with $x^2 + y^2 = 9$?

ANSWER None — $D = -76$

3. For which c is $y = 2x + c$ a tangent to $x^2 + y^2 = 20$?

ANSWER $c = \pm 10$

4. For which c is $y = x + c$ a tangent to $x^2 + y^2 = 25$?

ANSWER $c = \pm 5\sqrt{2}$

5. Where does $x = 4$ meet $x^2 + y^2 = 25$?

ANSWER $(4, 3), (4, -3)$

6. Chord of $x^2 + y^2 = 100$ along $y = 6$

ANSWER 16

7. Where does $y = 2x$ meet $(x - 5)^2 + y^2 = 5$?

ANSWER $(1, 2), (4, 8)$

Hard · stretch and reason

7 PROBLEMS

1. Length of the chord of $x^2 + y^2 = 25$ on $y = x + 1$

ANSWER $7\sqrt{2} \approx 9.90$

2. The tangents from $(0, 10)$ to $x^2 + y^2 = 25$

ANSWER $y = \pm\sqrt{3}x + 10$

3. For which k does $y = kx$ touch $(x - 5)^2 + y^2 = 9$?

ANSWER $k = \pm\frac{3}{4}$

4. Where do $x^2 + y^2 = 25$ and $(x - 8)^2 + y^2 = 25$ meet?

ANSWER $(4, 3), (4, -3)$

5. Shortest distance from $(8, 6)$ to $x^2 + y^2 = 25$

ANSWER 5

6. The line $3x + 4y = 25$ and the circle $x^2 + y^2 = 25$: how many points?

ANSWER One — it is the tangent at $(3, 4)$

7. For which r does $y = x + 6$ touch $x^2 + y^2 = r^2$?

ANSWER $r = 3\sqrt{2}$

07 Homework

Homework — 5 tasks

Always give both coordinates of every point you find.

- Find where $y = 2x - 5$ meets $x^2 + y^2 = 25$.
- Show that $y = x + 8$ does not meet $x^2 + y^2 = 25$, using the discriminant.
- Find the values of c for which $y = 3x + c$ is a tangent to $x^2 + y^2 = 40$.
- Find the length of the chord cut on $x^2 + y^2 = 169$ by the line $y = 5$.

5. Find where the circles $x^2 + y^2 = 25$ and $(x - 6)^2 + y^2 = 25$ intersect.